

ARUN ARUNACHALAM

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PROFESSIONAL SUMMARY

Technology executive with 17+ years engineering production-grade analytics, pricing, risk, and data platforms across capital markets at Bank of America. Deep expertise in designing and delivering scalable, cloud-native analytics capabilities—spanning cross-asset pricing, structured products, real-time risk metrics, and AI/ML-driven predictive analytics—aligned to the full data and analytics engineering lifecycle. Proven record modernizing legacy infrastructure into event-driven, distributed systems that process \$1T+ daily and enable measurable business outcomes at enterprise scale. Combines hands-on architectural depth in Spark, Databricks, Kafka, and AWS with applied GenAI (RAG pipelines, agentic AI) to accelerate delivery of intelligent, production-ready analytics products.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

Analytics Engineering & Pricing Platforms | AI/ML-Driven Predictive Analytics (RAG, Agentic AI) | Cross-Asset Risk & Derived Metrics | Cloud-Native Architecture (AWS, Kubernetes) | Event-Driven Data Systems (Kafka) | Data Engineering (Spark, Databricks, ETL/ELT) | Production-Grade Platform Delivery | Structured & Private Markets Technology | Regulatory & Compliance Systems | Executive Stakeholder Leadership

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Owned end-to-end engineering of real-time pricing, risk, and derived analytics platforms supporting **\$1T+ daily processing** across cross-asset trading systems
- Designed and delivered scalable, production-grade analytics capabilities on cloud-native, event-driven architecture (AWS, Kafka, Kubernetes), improving throughput by **40%**
- Modernized legacy data infrastructure into distributed, high-availability analytics systems with **99.97% uptime** across 50+ integrated platforms
- Implemented AI/ML-driven anomaly detection and AI-assisted development workflows, accelerating delivery velocity by **15%**
- Ensured regulatory compliance (FRB, FINRA, SEC) across mission-critical analytics and risk systems
- Led global engineering teams (**30+ across US/UK/India**) delivering enterprise-scale analytics transformation

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Engineered real-time pricing and analytics platforms for structured products and mortgage trading, contributing to **~\$2B capital efficiency** through risk system consolidation

- Built derived metrics and analytics capabilities supporting complex, multi-asset pricing and risk workflows
- Delivered production-grade analytics infrastructure for structured and private markets trading desks

Bank of America — AVP, Whole Loan Trading Desk Support (*prior to 2011*)

- Architected and delivered end-to-end Java EE trading analytics and support applications for whole loan trading
- Introduced CI/CD practices reducing release cycle from monthly to weekly, improving analytics delivery cadence

Senior Programmer Analyst — Knight Trading (*2004*)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (*2003*)

Sr. Java Web Developer — Mellon Investor Services (*2001*)

Sr. Java GUI Developer — Thomson Financials (*2001*)

Sr. Java GUI Developer — Icor Brokerage (*2000*)

AI & INNOVATION

- Building GenAI platforms (RAG pipelines, agentic AI systems) for real-world analytics and operational applications
- Applied ML for anomaly detection and operational intelligence across capital markets data systems
- Driving AI-assisted SDLC workflows to improve delivery speed, quality, and predictive analytics capabilities

EDUCATION

Bachelor of Technology — Anna University

SKILLS

Architecture: Cloud-native architecture (AWS, Kubernetes), event-driven systems (Kafka), distributed systems, enterprise analytics architecture, high-availability platforms (99.97% uptime) **AI & Data:** GenAI (RAG pipelines, agentic AI), ML anomaly detection, predictive analytics, AI-assisted SDLC, Spark, Databricks, ETL/ELT, Snowflake, Hadoop, Cassandra, MongoDB **Engineering:** Java, Python, Node.js, SQL, Oracle, Sybase, REST/SOA, WebLogic, Kafka, Kubernetes, AWS **Leadership:** Executive stakeholder management, global team leadership (30+ engineers), P&L ownership, strategic planning, program management, SDLC, Agile, ITIL, SOX, FRB/FINRA/SEC compliance **Tools:** ServiceNow, JIRA, Git, Visio, Microsoft Project